Wider and deeper without mitigation: IBM's dual-unitary chain at 91 qubits, and three-dimensional lattices through a placement-only OS

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Abstract

Paper 7 of this series carried user-written two-dimensional circuits through an operating layer (psf-os) that reads no calibration-quality number, adds its own controls to each job and applies no error mitigation. Here the same policy is carried to a chain that no fixed shape can hold and to three-dimensional lattices that the heavy-hex coupling graph cannot embed. IBM's dual-unitary kicked-Ising chain (arXiv:2411.00765) at its largest width, $N = 91$, read raw on ibm_marrakesh: the decay rates agree with $\cos 2h$ at all three h (largest $|k - \cos 2h|$ 0.0037) and every depth up to $t = 24$ (2 205 CZ) is readable, on a chain placed by a fixed rule round the failed elements; a second run on ibm_fez, with the ruler $h = 0$ in both jobs, again found all three rates in agreement, while $h = 0.15$ sat $+0.0037$ above $\cos 2h$ on both devices. In three dimensions, a place rule that judges the place with the highest own mirror replaced a gate that had refused every place at 48 and 72 sites; kicked Ising on $4 \times 4 \times 3$ read at every angle and depth, Heisenberg XXX on $4 \times 2 \times 2$ read with a Néel retention of 0.72 to 0.51. Two new layouts cut the transport CZ of the Heisenberg step by 42%, and were compared with the old one on the same places of two devices; an Ising layout with 46% fewer CZ came closer to the exact answer on $4 \times 2 \times 2$ but farther on $4 \times 3 \times 2$. On the way we changed our main measure for such comparisons, from the retention of an order parameter to a normalised per-site error, because noise that slows the dynamics can raise the retention. Pair creation in the Schwinger model read in 16 of 18 circuits and beat both trivial guesses in all 18. A light-cone test failed as predicted: every front arrived late, while the shape of the spread was reproduced. Every device result re-scores to the same numbers from its committed raw counts (23 of 23). No quantum advantage is claimed: every answer is computed exactly on a classical machine.

1 Introduction

Paper 7 (doi:10.5281/zenodo.24046673) reported user-written circuits on the two-dimensional lattice, a daily record and IBM's dual-unitary chain at 51 and 71 sites, all under one policy: placement from the coupling graph and the failed elements only, the device chosen by its queue, the job's own controls (floor, mirror, ring sentinel) in the same submission, no mitigation, answers sealed by hash before each run. This paper keeps the policy unchanged and asks whether it carries to

- a wider chain, where no fixed shape fits on the day's records and a fixed walk round the failed elements (rule C) must place it;
- three-dimensional lattices, which heavy-hex cannot embed, so that the OS must fold them and move qubits.

What is new in the OS: rule C for chains, a 3D layer with fold and line layouts, the place rule "top", and a change of the main measure for layout comparisons (section 6). Not claimed: quantum advantage (every answer is computed classically); that any mitigation works; that one layout is better in general (each comparison is per place and day, with the measure fixed in advance).

2 The dual-unitary chain at $N = 91$

2.1 Placement

On the records of 7 to 9 November no fixed chain shape of daily/du91 fits 91 sites on any of the three devices. Rule C (`psf_os.chain.place`, adopted on 6 November) walks round the failed elements in a fixed neighbour order from fixed starts and takes the first fit; it is not a search, and the steps are recorded. On ibm_marrakesh on 9 November it placed 91 sites in 1 665 steps. Two checks preceded the run: the compiled circuits at $h = 0$ (Clifford) were simulated at full size by the stabilizer method ($|\langle X \rangle| = 1$ on the light-cone site at every t), and the closed form for $h > 0$ had been checked against statevectors and an MPS.

2.2 Result (declared verdicts)

`results/du91-2026-11-09`, prereg/du91_2026-11-09.md. READS (MSE 0.0393 against a mean square 0.699). Decay rates $k = 0.9956$, 0.9821, 0.9590 against $\cos 2h = 0.9950$, 0.9801, 0.9553: AGREES at all three h. Readable depth 24 for every h. Sentinels: job A DEAD, job B CLEAN (as recorded). Table 1 sets the three widths side by side.

2.3 After the fact

The scatter of the rate fit is larger than shot noise ($\chi^2/5 = 38$, 26, 19 for the three h; with it the deviations are 0.4 to 1.7 σ; `notes/du91_2026-11-09`). In all three widths $h = 0.15$ came out slower than $\cos 2h$, and in all three the ruler ($h = 0$) and $h = 0.15$ sat in different jobs.

2.4 Second run, with the ruler in every job

`results/night-2026-11-10/du91r`, prereg/du91r_2026-11-10.md. On ibm_fez (the declared tie rule among fez and kingston), rule C placed 91 sites; job A held $h = 0$, 0.05, 0.10 and job B $h = 0$, 0.15, and every rate was divided by the ruler of its own job. READS (MSE 0.287 against 0.760). $k = 0.9960$, 0.9814, 0.9590 against 0.9950, 0.9801, 0.9553: AGREES at all three h; readable depth 24 (16 for $h = 0.10$). Beside: the two rulers agree (ratio 0.998 ± 0.0015), so the jobs saw the device alike, and $h = 0.15$ is again $+0.0037$ above $\cos 2h$, the same as on marrakesh. The earlier slowness of $h = 0.15$ is therefore not a job-to-job difference; in this run it is inside two standard errors. The signal was weaker than on marrakesh: every series sat near 0.04 to 0.11 at $t = 20$–24 whatever its exact value (Table 1).

3 The three-dimensional layer of the OS

3.1 Layouts

A lattice $L \times M \times N$ is folded onto heavy-hex (fold v3.1): blocks are placed by translation and the bonds between them are carried by swaps over bridge qubits, on fixed schedules found offline and kept as constants. Two layouts followed: *carry* (an x bond waits for a later swap of the same step and rides on it) and *line* (two blocks interleaved on one line, no empty swap, even steps reversed, the y bonds and line operations at step boundaries merged). Table 4 gives the CZ per step.

3.2 Choosing the place

A pre-run gate (probes for floor marks and low mirror sites) worked at 16 sites: on 7 November on ibm_marrakesh it passed one of four places, which was judged (`results/os3d-gate-2026-11-07`). At 48 sites it refused every one of five places on ibm_fez (all had $|1\rangle$ floor marks), so no 3D arm was judged (`results/os3d-mid-2026-11-07`). The owner replaced the stop by the rule "top": the place whose own $t = 1$ mirror is highest is judged, and the gate's marks are recorded beside. On ibm_kingston the judged $4 \times 4 \times 3$ place read at $t = 1$–3 (retention 0.92, 0.95, 0.95) and had the smallest error of the five places; the gate alone would have refused it and every other place. $4 \times 6 \times 3$ (72 sites, one place) read with retention 0.88, 0.88, 0.86 (`results/os3d-kingston-top-2026-11-07`).

4 Physics A: kicked Ising on 3D lattices

`results/os3d-physA-2026-11-07`, ibm_fez, rule top, $\theta = 0.4, 0.6, 0.8$. $4 \times 4 \times 3$ (48 sites, CZ up to 1 464) READS at every θ and t = 1–3, with retention 0.73 to 0.83; $4 \times 2 \times 2$ READS at every θ and t = 1–6 (Table 2). The magnetisation DIFFERS on most rows under the absolute line of 0.05; as in paper 7, the retention beside it shows how much signal was kept.

5 Physics B: Heisenberg XXX on $4 \times 2 \times 2$

`results/os3d-heisB-2026-11-08`, ibm_fez. READS at t = 1–4, DIFFERS; Néel retention 0.72, 0.62, 0.51, 0.53 at 176 to 692 CZ (Table 3).

6 Layouts compared on the same places

6.1 Transport CZ

Old, carry and line give 176, 152 and 116 CZ at $t = 1$ and 692, 596 and 404 at $t = 4$ for the Heisenberg step (42% fewer at $t = 4$ with line); for Ising, line gives 60 against fold's 108 per step at $4 \times 2 \times 2$ and 126 against 234 at $4 \times 3 \times 2$ (Table 4). Every layout was checked against the exact answer before any run.

6.2 Declared comparison

The three Heisenberg layouts ran in one job at the same places on ibm_marrakesh and on ibm_kingston (8 November; Table 5). The declared measure was the difference of Néel retention, line minus old, with an expectation of ≥ 0. At the judged place on marrakesh: +0.039, +0.079, +0.096, +0.077 at t = 1–4 (resolved at $t = 2$ and 3). On kingston: +0.022, +0.056, +0.050, −0.051; the expectation failed at $t = 4$.

6.3 Why the measure changed

A retention can rise for the wrong reason: noise that slows the exchange keeps the Néel order longer. From 8 November the main measure for layout comparisons is the normalised per-site error nMSE $= \overline{(z-e)^2}/\overline{e^2}$, with the retention beside it (`notes/reporting/layout_comparison_metric.md`). Applied after the fact to the same arms, line has the lower nMSE in 28 of 32 place-and-t cells (Table 5). The two runs keep their declared verdicts.

6.4 Ising: line against fold

`results/night-2026-11-10/isingL`, prereg/isingL_2026-11-09.md, nMSE as the declared measure. The declared order gave ibm_marrakesh, where the job's ring sentinel could not be placed; by the owner's decision the next device, ibm_fez, was used (recorded before the run). At the judged $4 \times 2 \times 2$ place line minus fold was −0.0007, −0.0022, −0.0082, −0.0244 at t = 1–4 (resolved at $t = 3$ and 4); at the judged $4 \times 3 \times 2$ place +0.0026, +0.0772, +0.0812, +0.0439 (resolved at t = 2–4). The prediction (line closer everywhere) held on $4 \times 2 \times 2$ and failed on $4 \times 3 \times 2$, where line was farther in all 8 place-and-t cells with 46% fewer CZ (Table 9). Fewer CZ did not give a closer answer there; this run does not say why.

7 A new problem: pair creation in the Schwinger model

`results/schw-2026-11-09`, ibm_marrakesh; Martinez et al. (2016) form, $N = 6$ and 8, parameter sets M1–M3, 1, 2 and 5 Trotter steps, a swap network on a chain. READS 16 of 18, BEATS (both the bare vacuum and the all-zero guess) 18 of 18, particle density AGREES within 0.05 in 14 of 18. The two that did not read are $N = 8$ at 5 steps for M1 and M3 (420 CZ). The total charge $\sum Z$ (exactly 0) is recorded raw, never used to correct; it drifted to +0.1 to +0.6 (Table 6). An estimate after the run (`notes/schwinger_scale_2026-11-10`): the swap network costs $3(N-1)$ CZ per qubit per step, so larger N would not read with this circuit.

8 The light cone in 3D, retried on a device

`results/lc3d-2026-11-09`, ibm_marrakesh: one flipped spin in the all-up $4 \times 2 \times 2$ XXX lattice, with a no-flip arm in the same job to subtract the background. The test had been dropped on an estimate that noise flips would bury the signal; it was run after all, with the failure beyond distance 1 predicted. Every front DIFFERS at the judged place: distance 1 arrived at $t = 2$ (exact $t = 1$), distance 2 at $t = 6$ (exact 3), distance 3 not within $t = 6$. The background held 2.1 to 3.9 flipped spins and the excitation kept 0.76 down to 0.43 of itself, while the shape of the spread matched the exact one within a total variation distance of 0.025 to 0.144 (Table 8).

9 The simulator and the device

A trajectory simulator with that day's calibrated noise, never fitted to the device (`notes/os3d_sim_2026-11-07/realsim.py`), was compared once with the device (`notes/simproto_2026-11-08`). For kicked Ising $4 \times 2 \times 2$ the simulated value over the device value had a mean of 0.987 (0.929 to 1.025); for Heisenberg it overestimated the retention by 15% to 28% (Table 7). The difference between layouts, not the level, is what the comparisons above need.

10 Limits and lessons

- *Two mirrors* (a second, twirled mirror) were not added: a planted error of 0.02 could not be told apart within 1.5 times the device time; a hypothesis about coherent CZ errors was withdrawn after a computing error was found (`notes/two_mirrors_2026-11-08`).
- *Three mirrors* usable at $t = 1$ only; at $t \geq 2$ the product rule they rest on fails under gate-local noise alone (`notes/three_mirrors_2026-11-09`).
- The retention bias that led to nMSE (section 6); a declared expectation that failed (kingston, $t = 4$); the fit scatter at $N = 91$ and the $+0.0037$ of $h = 0.15$, the same on two devices (section 2).
- An Ising layout with 46% fewer CZ was farther from the exact answer on $4 \times 3 \times 2$: the CZ count is not the whole cost; and on 10 November the declared job could not be built on ibm_marrakesh (its ring sentinel did not land on a record with 14 failed qubits).
- Three devices of one family and a few days; the 3D lattices are small (≤ 72 sites) and every answer is classically computable.

11 Reproduction

`python reproduce.py` re-runs every committed scorer, unchanged, on a copy of each result from its committed raw counts and compares every number with the committed scores. For the 23 device results of papers 7 and 8 it prints SAME for all 23; for three Heisenberg results whose scorer gained keys later, the scorer as committed at the time is taken from the history. Answers files are regenerated by committed code and their hashes compared with those committed before each run; job folders and job identifiers are committed.

12 Conclusion

The no-map, no-mitigation policy carried to IBM's 91-site dual-unitary chain, placed by a fixed walk round failed elements, with decay rates within 0.004 of $\cos 2h$ on two devices; and to 3D lattices up to 72 sites through a fold of heavy-hex, with the place chosen by its own mirror. Layouts that cut the transport CZ by 42% were compared on the same places of two devices, and the comparison taught us to measure per-site error rather than retention; for Ising, fewer CZ helped on one lattice and not on the next. A new problem read, and a light-cone test failed as predicted. No advantage is claimed.

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Data and code

TOQB (`https://github.com/TN-Holdings-LLC/TOQB`) holds the declarations (`prereg/`), the results with raw counts, job identifiers and released answers (`results/`), the scorers (`gpu/`), `reproduce.py` and the tables of this paper (`notes/paper8/tables.py`). psf-os (`https://github.com/TN-Holdings-LLC/psf-os`) is the operating layer. [**PENDING: commit hashes at release.**]

References

1. IBM Quantum, dual-unitary kicked-Ising experiment at 51, 71 and 91 qubits, arXiv:2411.00765.
2. E. A. Martinez et al., Real-time dynamics of lattice gauge theories with a few-qubit quantum computer, Nature 534, 516 (2016), arXiv:1605.04570.
3. TN Holdings, paper 7 of this series, doi:10.5281/zenodo.24046673.
4. TN Holdings, paper 6 of this series, doi:10.5281/zenodo.23883938.

A Generated tables

Generated by `notes/paper8/tables.py` from the committed records; no number in them is typed by hand.

Table 1: Table 1. The dual-unitary chain (arXiv 2411.00765) raw: N = 51, 71 (paper 7) and 91 twice (results/du51-2026-11-07, du71-2026-11-07, du91-2026-11-09, night-2026-11-10/du91r; in the second run each rate is against the ruler of its own job). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

N	device	placement	CZ (t 24)	READS (MSE / mean sq.)	k - cos2h, h 0.05	h 0.10	h 0.15	readable depth (h 0 / .05 / .10 / .15)
51	ibm_kingston	shape S1	1225	READS (0.0403 / 0.699)	+0.0000 AGREES	+0.0020 AGREES	+0.0045 AGREES	24 / 24 / 24 / 24
71	ibm_kingston	shape S1	1715	READS (0.0508 / 0.699)	-0.0002 AGREES	-0.0002 AGREES	+0.0017 AGREES	24 / 24 / 24 / 24
91	ibm_marrakesh	C	2205	READS (0.0393 / 0.699)	+0.0006 AGREES	+0.0020 AGREES	+0.0037 AGREES	24 / 24 / 24 / 24
91	ibm_fez (2nd run, ruler in both jobs)	C	2205	READS (0.2872 / 0.760)	+0.0010 AGREES	+0.0013 AGREES	+0.0037 AGREES	24 / 24 / 16 / 24

Table 2: Table 2. Physics A: kicked Ising on 3D lattices, ibm_fez 2026-11-07, the judged place (rule top), 3D sites (results/os3d-physA-2026-11-07). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

lattice	theta	t	CZ	M	M exact	retention	READS	AGREES	BEATS
4x4x3	0.4	1	488	0.729	0.921	0.791	READS	DIFFERS	DOES NOT BEAT
4x4x3	0.4	2	976	0.638	0.811	0.787	READS	DIFFERS	DOES NOT BEAT
4x4x3	0.4	3	1464	0.534	0.729	0.733	READS	DIFFERS	BEATS
4x4x3	0.6	1	488	0.659	0.825	0.798	READS	DIFFERS	DOES NOT BEAT
4x4x3	0.6	2	976	0.503	0.628	0.800	READS	DIFFERS	BEATS
4x4x3	0.6	3	1464	0.393	0.512	0.767	READS	DIFFERS	BEATS
4x4x3	0.8	1	488	0.556	0.697	0.798	READS	DIFFERS	BEATS
4x4x3	0.8	2	976	0.367	0.440	0.834	READS	DIFFERS	BEATS
4x4x3	0.8	3	1464	0.255	0.312	0.818	READS	DIFFERS	BEATS
4x2x2	0.4	1	108	0.829	0.921	0.900	READS	DIFFERS	DOES NOT BEAT
4x2x2	0.4	2	216	0.700	0.794	0.882	READS	DIFFERS	BEATS
4x2x2	0.4	3	324	0.616	0.717	0.859	READS	DIFFERS	BEATS
4x2x2	0.4	4	432	0.548	0.634	0.863	READS	DIFFERS	BEATS
4x2x2	0.4	5	540	0.463	0.537	0.862	READS	DIFFERS	BEATS
4x2x2	0.4	6	648	0.408	0.492	0.830	READS	DIFFERS	BEATS
4x2x2	0.6	1	108	0.744	0.825	0.901	READS	DIFFERS	BEATS
4x2x2	0.6	2	216	0.544	0.607	0.895	READS	DIFFERS	BEATS
4x2x2	0.6	3	324	0.450	0.513	0.877	READS	DIFFERS	BEATS
4x2x2	0.6	4	432	0.380	0.448	0.848	READS	DIFFERS	BEATS
4x2x2	0.6	5	540	0.326	0.418	0.779	READS	DIFFERS	BEATS
4x2x2	0.6	6	648	0.284	0.429	0.661	READS	DIFFERS	BEATS
4x2x2	0.8	1	108	0.630	0.697	0.904	READS	DIFFERS	BEATS
4x2x2	0.8	2	216	0.390	0.425	0.918	READS	AGREES	BEATS
4x2x2	0.8	3	324	0.299	0.321	0.932	READS	AGREES	BEATS
4x2x2	0.8	4	432	0.232	0.304	0.764	READS	DIFFERS	BEATS
4x2x2	0.8	5	540	0.188	0.301	0.625	READS	DIFFERS	BEATS
4x2x2	0.8	6	648	0.152	0.278	0.547	READS	DIFFERS	BEATS

Table 3: Table 3. Physics B: Heisenberg XXX 4x2x2, ibm_fez 2026-11-08, the judged place 0 (results/os3d-heisB-2026-11-08/fez). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

t	CZ	m_s	m_s exact	retention	nMSE (after the fact)	READS	AGREES
1	176	0.669	0.932	0.719	0.088	READS	DIFFERS
2	348	0.469	0.757	0.619	0.163	READS	DIFFERS
3	520	0.270	0.530	0.509	0.299	READS	DIFFERS
4	692	0.164	0.308	0.532	0.369	READS	DIFFERS

Table 4: Table 4. CZ per layout and t (Heisenberg: heisE marrakesh build, judged place; Ising: the isingL build of the noiseless check, place 0; notes/isingL_noiseless_2026-11-10). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

model	layout	CZ t 1	t 2	t 3	t 4
Heisenberg 4x2x2	old	176	348	520	692
Heisenberg 4x2x2	carry	152	300	448	596
Heisenberg 4x2x2	line	116	212	308	404
Ising 4x2x2	fold	108	216	324	432
Ising 4x2x2	line	60	120	180	240
Ising 4x3x2	fold	234	468	702	936
Ising 4x3x2	line	126	252	378	504

Table 5: Table 5. Heisenberg layouts on the same places: line against old (results/os3d-heisE-2026-11-08, os3d-heisE-kingston-2026-11-08); the declared measure was the retention difference, nMSE is after the fact (notes/reporting/layout_comparison_metric.md). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

device	place	t	CZ old / line	retention line - old (declared)	resolved	nMSE old	nMSE line (after the fact)
ibm_marrakesh	0	1	176 / 116	+0.038	no	0.083	0.064
ibm_marrakesh	0	2	348 / 212	+0.062	yes	0.154	0.100
ibm_marrakesh	0	3	520 / 308	-0.016	no	0.261	0.270
ibm_marrakesh	0	4	692 / 404	+0.009	no	0.449	0.322
ibm_marrakesh	1 (judged)	1	176 / 116	+0.039	no	0.046	0.031
ibm_marrakesh	1 (judged)	2	348 / 212	+0.079	yes	0.118	0.075
ibm_marrakesh	1 (judged)	3	520 / 308	+0.096	yes	0.229	0.132
ibm_marrakesh	1 (judged)	4	692 / 404	+0.077	no	0.425	0.267
ibm_marrakesh	2	1	176 / 116	+0.037	no	0.055	0.042
ibm_marrakesh	2	2	348 / 212	+0.057	no	0.124	0.091
ibm_marrakesh	2	3	520 / 308	+0.094	yes	0.228	0.152
ibm_marrakesh	2	4	692 / 404	+0.111	no	0.321	0.236
ibm_marrakesh	3	1	212 / 128	+0.012	no	0.138	0.128
ibm_marrakesh	3	2	420 / 236	+0.070	yes	0.193	0.130
ibm_marrakesh	3	3	628 / 344	-0.053	no	0.377	0.437
ibm_marrakesh	3	4	836 / 452	-0.037	no	0.344	0.518
ibm_kingston	0	1	176 / 116	+0.029	no	0.046	0.034
ibm_kingston	0	2	348 / 212	+0.044	no	0.118	0.091
ibm_kingston	0	3	520 / 308	+0.000	no	0.223	0.212
ibm_kingston	0	4	692 / 404	-0.000	no	0.429	0.409
ibm_kingston	1 (judged)	1	176 / 116	+0.022	no	0.043	0.035
ibm_kingston	1 (judged)	2	348 / 212	+0.056	no	0.128	0.090
ibm_kingston	1 (judged)	3	520 / 308	+0.050	no	0.250	0.186
ibm_kingston	1 (judged)	4	692 / 404	-0.051	no	0.466	0.455
ibm_kingston	2	1	176 / 116	+0.057	yes	0.049	0.026
ibm_kingston	2	2	348 / 212	+0.109	yes	0.142	0.070
ibm_kingston	2	3	520 / 308	+0.144	yes	0.302	0.151
ibm_kingston	2	4	692 / 404	+0.109	no	0.471	0.394
ibm_kingston	3	1	176 / 116	+0.080	yes	0.135	0.056
ibm_kingston	3	2	348 / 212	+0.057	no	0.233	0.183
ibm_kingston	3	3	520 / 308	+0.096	yes	0.366	0.232
ibm_kingston	3	4	692 / 404	+0.001	no	0.388	0.444

Table 6: Table 6. Schwinger pair creation from the bare vacuum, ibm_marrakesh 2026-11-09, sentinel CLEAN (results/schw-2026-11-09). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

N	set	steps	CZ	nMSE	READS	BEATS	nu / exact	nu	Sum Z (exact 0)
6	M1	1	45	0.009	READS	BEATS	0.133 / 0.097	AGREES	+0.124
6	M1	2	90	0.029	READS	BEATS	0.332 / 0.315	AGREES	+0.230
6	M1	5	225	0.446	READS	BEATS	0.443 / 0.333	DIFFERS	+0.406
6	M2	1	45	0.013	READS	BEATS	0.140 / 0.097	AGREES	+0.154
6	M2	2	90	0.020	READS	BEATS	0.331 / 0.327	AGREES	+0.214
6	M2	5	225	0.188	READS	BEATS	0.574 / 0.563	AGREES	+0.418
6	M3	1	45	0.008	READS	BEATS	0.130 / 0.097	AGREES	+0.115
6	M3	2	90	0.022	READS	BEATS	0.333 / 0.318	AGREES	+0.215
6	M3	5	225	0.384	READS	BEATS	0.446 / 0.361	DIFFERS	+0.367
8	M1	1	84	0.015	READS	BEATS	0.146 / 0.101	AGREES	+0.206
8	M1	2	168	0.036	READS	BEATS	0.345 / 0.328	AGREES	+0.370
8	M1	5	420	0.677	DOES NOT READ	BEATS	0.467 / 0.328	DIFFERS	+0.612
8	M2	1	84	0.013	READS	BEATS	0.144 / 0.101	AGREES	+0.186
8	M2	2	168	0.046	READS	BEATS	0.343 / 0.339	AGREES	+0.369
8	M2	5	420	0.234	READS	BEATS	0.569 / 0.557	AGREES	+0.532
8	M3	1	84	0.016	READS	BEATS	0.148 / 0.101	AGREES	+0.259
8	M3	2	168	0.044	READS	BEATS	0.344 / 0.331	AGREES	+0.366
8	M3	5	420	0.566	DOES NOT READ	BEATS	0.460 / 0.359	DIFFERS	+0.539

Table 7: Table 7. The simulator (notes/os3d_sim_2026-11-07/realsim.py –expect; noise from the device's calibration of that day, never fitted) against the device (notes/simproto_2026-11-08). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

model	arm	CZ	exact	device	simulator (+- SE)	simulator / device retention
Heisenberg B	h3d0_t1	176	0.932	0.669	0.771 +- 0.009	1.152
Heisenberg B	h3d0_t2	348	0.757	0.469	0.551 +- 0.009	1.176
Heisenberg B	h3d0_t3	520	0.530	0.270	0.345 +- 0.009	1.278
Heisenberg B	h3d0_t4	692	0.308	0.164	0.196 +- 0.008	1.198
Ising A 4x2x2	g3d3_th04_t1	108	0.921	0.829	0.849 +- 0.006	1.025
Ising A 4x2x2	g3d3_th04_t2	216	0.794	0.700	0.704 +- 0.007	1.006
Ising A 4x2x2	g3d3_th04_t3	324	0.717	0.616	0.622 +- 0.007	1.010
Ising A 4x2x2	g3d3_th04_t4	432	0.634	0.548	0.539 +- 0.009	0.984
Ising A 4x2x2	g3d3_th04_t5	540	0.537	0.463	0.463 +- 0.008	1.002
Ising A 4x2x2	g3d3_th04_t6	648	0.492	0.408	0.408 +- 0.008	1.000
Ising A 4x2x2	g3d3_th06_t1	108	0.825	0.744	0.755 +- 0.005	1.015
Ising A 4x2x2	g3d3_th06_t2	216	0.607	0.544	0.555 +- 0.005	1.021
Ising A 4x2x2	g3d3_th06_t3	324	0.513	0.450	0.454 +- 0.006	1.009
Ising A 4x2x2	g3d3_th06_t4	432	0.448	0.380	0.384 +- 0.006	1.009
Ising A 4x2x2	g3d3_th06_t5	540	0.418	0.326	0.306 +- 0.006	0.939
Ising A 4x2x2	g3d3_th06_t6	648	0.429	0.284	0.264 +- 0.006	0.929
Ising A 4x2x2	g3d3_th08_t1	108	0.697	0.630	0.637 +- 0.005	1.012
Ising A 4x2x2	g3d3_th08_t2	216	0.425	0.390	0.389 +- 0.003	0.999
Ising A 4x2x2	g3d3_th08_t3	324	0.321	0.299	0.286 +- 0.004	0.957
Ising A 4x2x2	g3d3_th08_t4	432	0.304	0.232	0.225 +- 0.005	0.968
Ising A 4x2x2	g3d3_th08_t5	540	0.301	0.188	0.179 +- 0.005	0.952
Ising A 4x2x2	g3d3_th08_t6	648	0.278	0.152	0.141 +- 0.004	0.930

Table 8: Table 8. The light cone of one flipped spin, 4x2x2 XXX, ibm_marrakesh 2026-11-09, judged place 0; fronts: d 1: exact t 1, measured 2 (FRONT DIFFERS); d 2: exact t 3, measured 6 (FRONT DIFFERS); d 3: exact t 6, measured none (FRONT DIFFERS) (results/lc3d-2026-11-09). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

t	CZ	excitation kept K	background flips	shape TVD
1	116	0.758	2.317	0.025
2	212	0.646	2.147	0.056
3	308	0.614	2.879	0.049
4	404	0.572	3.099	0.053
5	500	0.441	3.621	0.144
6	596	0.432	3.876	0.102

Table 9: Table 9. Ising (theta 0.6): the line layout against the fold at the same places, ibm_fez 2026-11-10; main measure nMSE as declared (results/night-2026-11-10/isingL). From the committed records named in each caption; raw device values, no mitigation; verdicts as declared before each run; after-the-fact numbers are marked so.

lattice	place	t	CZ fold / line	nMSE fold	nMSE line	line - fold	resolved
4x2x2	0	1	108 / 60	0.0149	0.0141	-0.0008	no
4x2x2	0	2	216 / 120	0.0239	0.0300	+0.0060	yes
4x2x2	0	3	324 / 180	0.0647	0.0944	+0.0297	yes
4x2x2	0	4	432 / 240	0.1045	0.1498	+0.0454	yes
4x2x2	1	1	108 / 60	0.0191	0.0118	-0.0074	yes
4x2x2	1	2	216 / 120	0.0227	0.0175	-0.0051	yes
4x2x2	1	3	324 / 180	0.0954	0.0381	-0.0574	yes
4x2x2	1	4	432 / 240	0.1852	0.0776	-0.1076	yes
4x2x2	2	1	108 / 60	0.0169	0.0191	+0.0022	no
4x2x2	2	2	216 / 120	0.0295	0.0287	-0.0008	no
4x2x2	2	3	324 / 180	0.0581	0.0476	-0.0105	yes
4x2x2	2	4	432 / 240	0.1001	0.0609	-0.0392	yes
4x2x2	3 (judged)	1	108 / 60	0.0145	0.0139	-0.0007	no
4x2x2	3 (judged)	2	216 / 120	0.0189	0.0166	-0.0022	no
4x2x2	3 (judged)	3	324 / 180	0.0406	0.0323	-0.0082	yes
4x2x2	3 (judged)	4	432 / 240	0.0732	0.0487	-0.0244	yes
4x3x2	0 (judged)	1	234 / 126	0.0332	0.0358	+0.0026	no
4x3x2	0 (judged)	2	468 / 252	0.0476	0.1248	+0.0772	yes
4x3x2	0 (judged)	3	702 / 378	0.1072	0.1884	+0.0812	yes
4x3x2	0 (judged)	4	936 / 504	0.2038	0.2477	+0.0439	yes
4x3x2	1	1	234 / 126	0.0334	0.0644	+0.0310	yes
4x3x2	1	2	468 / 252	0.0433	0.1173	+0.0740	yes
4x3x2	1	3	702 / 378	0.1349	0.2832	+0.1482	yes
4x3x2	1	4	936 / 504	0.2696	0.3897	+0.1201	yes